

Protecting a Factory from Cyberattacks Using Artificial Intelligence

Abstract — Industrial factories face increasing cyber threats targeting programmable logic controllers (PLCs), sensors, cameras, and human-machine interfaces (HMIs). This paper presents a distributed, real-time intrusion detection system (IDS) for smart factories, built on a three-tier Fog–Cloud–Dashboard architecture deployed via Docker. The Fog layer applies a tuned Decision Tree (DT v3, Recall = 99.89%) combined with an Isolation Forest anomaly detector to filter network traffic locally, forwarding only suspicious packets to the Cloud. The Cloud layer uses an XGBoost classifier (Accuracy = 76.15%) trained on the UNSW-NB15 dataset with SMOTE oversampling and adversarial augmentation to classify nine attack types, while SHAP values provide human-interpretable explanations for each decision. Two web dashboards serve distinct roles: a technical dashboard for security managers and a simplified bilingual (Arabic/English) observer interface. Live tests demonstrate a detection rate exceeding 97% under stress conditions, with sub-second response latency.

Keywords — Intrusion Detection, Fog Computing, XGBoost, SHAP Explainability, ICS Security

تواجه المصانع الذكية تهديدات إلكترونية متزايدة تستهدف المتحكمات المنطقية القابلة للبرمجة (PLC) والحساسات والكاميرات — **الملخص** وواجهات المشغل (HMI). تقدم هذه الورقة نظام كشف تسلل موزعاً وفي الوقت الحقيقي، مبنياً على معمارية ثلاثية الطبقات (Fog–Cloud–Dashboard) منشأة بواسطة Docker. تعمل طبقة الـ Fog باستخدام شجرة قرار محسنة (استرجاع 99.89%) مع كاشف شذوذ Isolation Forest لتصفية الحزم محلياً، وإحالة المشبوهة منها فحسب إلى السحابة. تُصنّف طبقة السحابة تسعة أنواع من الهجمات بواسطة XGBoost (دقة 76.15%) مع تفسير قرارات النموذج باستخدام SHAP. يتضمن النظام واجهتي ويب: الأولى للمدير التقني، والثانية للمراقب غير التقني بدعم ثنائي اللغة. أظهرت اختبارات التحميل الحي معدل كشف يتجاوز 97% مع زمن استجابة أقل من ثانية. كشف التسلل، الحوسبة الضبابية، XGBoost، قابلية التفسير، أمن الأنظمة الصناعية — **الكلمات المفتاحية**

1. INTRODUCTION

The convergence of operational technology (OT) and information technology (IT) has exposed industrial control systems to sophisticated cyberattacks. Incidents such as Stuxnet and Triton demonstrated that PLCs and SCADA systems are viable—and high-value—attack targets [1]. Traditional perimeter defenses are insufficient for heterogeneous factory environments where devices communicate over Modbus/TCP, MQTT, RTSP, and HTTP concurrently.

Fog computing brings computation closer to the data source, reducing latency and bandwidth consumption while enabling real-time decisions [2]. This work proposes an end-to-end IDS that places lightweight ML inference at the network edge (Fog) and delegates deeper analytics to a Cloud layer, achieving both speed and interpretability.

2. PROBLEM STATEMENT

Existing IDS solutions are predominantly cloud-centric, introducing latency incompatible with real-time OT requirements. Moreover, black-box models provide no actionable explanation to security operators, and most benchmarks are conducted on IT-oriented datasets that lack industrial protocol characteristics.

A further challenge is class imbalance: attack sub-types such as Worms and Shellcode constitute fewer than 0.05% of the UNSW-NB15 dataset, causing conventional classifiers to exhibit poor recall on rare but critical categories.

3. OBJECTIVES

The system targets the following measurable goals:

- Recall $\geq 99\%$ at the Fog binary filter to minimize missed attacks.

- Multi-class attack classification across nine categories with accuracy $> 75\%$.
- Sub-second end-to-end detection latency from packet arrival to dashboard alert.
- Human-interpretable SHAP explanations for every classified incident.
- Dual-audience dashboards: technical and non-technical.

4. PROPOSED SOLUTION

The architecture comprises five containerised services orchestrated by Docker Compose, communicating exclusively over HTTP on an isolated bridge network (factory_network). Figure 1 illustrates the data flow.

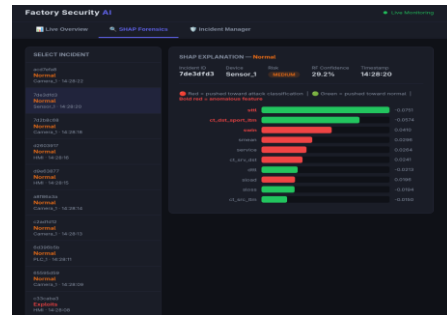


Figure 1. Technical Dashboard — Live Overview Tab

The Simulator generates synthetic network records every 0.5 s with a configurable 15% attack probability, drawing traffic profiles from attacks.json and devices.json. Packets are sent via HTTP POST to the Fog layer.

The Fog layer (port 5001) applies a two-stage filter: (i) Decision Tree v3 with class_weight {attack: 3×} and adversarial training data, and (ii) Isolation Forest trained exclusively on normal traffic to catch zero-day-like anomalies. A packet is escalated to Cloud if either model flags it.

The Cloud layer (port 5002) classifies escalated packets using XGBoost (300 estimators, max_depth 8), computes SHAP

values via TreeExplainer, records incidents to a daily CSV log, and pushes annotated alerts to the Dashboard.

5. INNOVATION & ORIGINALITY

Five distinguishing contributions:

- ① Dual-model Fog filter (DT + IsolationForest) achieving 99.89% recall with only 31 KB + 2 MB footprint — suitable for edge devices.
- ② Adversarial augmentation: three perturbation strategies (noise $\pm 40\%$, slowdown $\times 10$, padding) applied during training to harden the model against evasion attacks.
- ③ SMOTE minority oversampling targeting classes with fewer than 5,000 samples, addressing UNSW-NB15 imbalance.
- ④ SHAP forensics tab giving per-feature contribution bars with anomaly highlighting for operator understanding.
- ⑤ Role-based dual dashboard: security engineers see full forensics; floor observers see a simplified bilingual status card with a one-tap incident report button.

6. METHODOLOGY

6.1 Dataset

UNSW-NB15 [3] provides 175,341 training and 82,332 testing records spanning nine attack categories and one normal class. Raw features include 42 network flow attributes covering duration, byte counts, packet counts, TTL values, TCP flags, and connection-tracking counters.

6.2 Feature Engineering

Eight derived ratio features were appended: bytes_ratio, pkts_ratio, load_ratio,

duration_score, pkt_size_diff, jit_ratio, conn_intensity, and loss_rate. These amplify asymmetric traffic patterns characteristic of DoS and reconnaissance attacks, as confirmed by XGBoost importance scores (Figure 2).

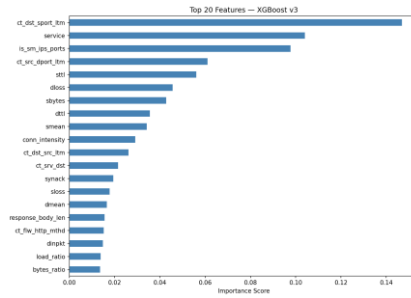


Figure 2. Top 20 Feature Importances — XGBoost v3

6.3 Training Pipeline

Training proceeded in three stages: (i) label encoding of categorical features (proto, service, state) with saved LabelEncoders for inference-time reuse; (ii) adversarial augmentation adding $3\times$ copies of attack records with noise, slowdown, and padding perturbations; (iii) SMOTE oversampling of minority classes to $\geq 5,000$ samples per class.

7. TOOLS & TECHNOLOGIES

Python 3.11 · scikit-learn · XGBoost · SHAP · imbalanced-learn · Flask · Chart.js · Docker / Docker Compose · UNSW-NB15 dataset

8. RESULTS

Table 1 compares model performance across training versions.

Table 1. Model Performance Comparison

Model	Role	Accuracy	Recall
Decision Tree v1	Fog – binary	84.45%	98.31%
Decision Tree v3	Fog – binary	81.42%	99.89%
Random Forest	Cloud – multi	76.69%	—
XGBoost v3	Cloud – multi	76.15%	—
IsolationForest	Fog – anomaly	—	≈62%

Figure 3 shows the DT v3 confusion matrix: only 52 attacks missed out of 45,332 (False Negative Rate = 0.11%), a 14× improvement over DT v1's 765 misses.

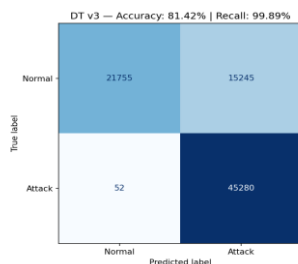


Figure 3. Decision Tree v3 — Confusion Matrix (Fog Layer)

Figure 4 presents the XGBoost v3 confusion matrix for nine-class classification. Generic (18,345) and Normal (28,011) classes are well-separated; rare classes (Worms: 24 correct) benefit from SMOTE augmentation.

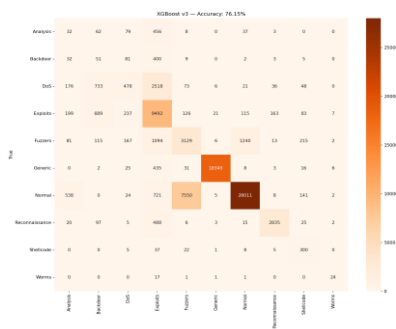


Figure 4. XGBoost v3 — Confusion Matrix (Cloud Layer)

Live stress tests with 480 mixed packets achieved a 97.7% detection rate (469/480) at a Fog filter rate of 2.3%, demonstrating

effective noise rejection without sacrificing attack capture.

This interface serves as the system's primary real-time monitoring dashboard, continuously refreshing to reflect the latest network activity. At the top, four statistical cards summarize the current state of the network: the total number of packets scanned by the Fog layer (480 packets), the number of normal packets filtered locally (11 packets), the number of detected attacks forwarded to the Cloud layer (469 attacks), and the overall detection rate, which reached 97.7%. Below the cards, a timeline chart plots the number of attacks moment by moment, enabling the security manager to track attack waves and peak periods with precision. Adjacent to it, a doughnut chart illustrates the distribution of attack types, including Backdoor, DoS, Fuzzers, and others. Beneath both charts, a Fog Node Health card displays operational details such as the total packets scanned, filtered packets, forwarded attacks, and the filtering rate. At the bottom of the page, a recent incidents table lists the latest recorded events, showing each incident's unique identifier, source device, attack type, risk level, RF confidence score, and current decision status.

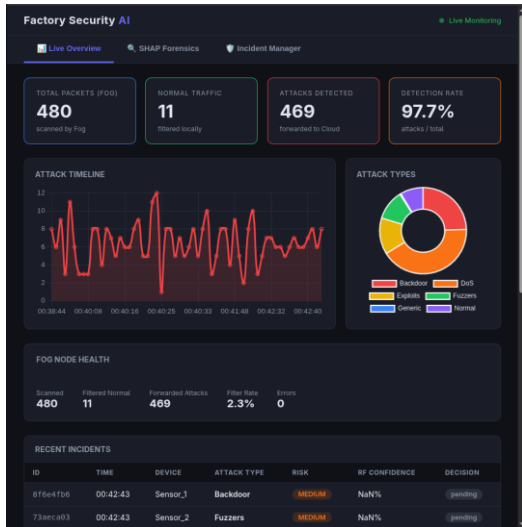


Figure 5. Technical Dashboard — Live Overview Tab

This interface represents the most analytically advanced component of the dashboard, addressing a fundamental question in AI-driven security systems: why did the model classify this packet as an attack? On the left panel, a scrollable list displays all recorded incidents, allowing the manager to select any entry for detailed forensic analysis. Upon selection, the right panel renders a full SHAP explanation for the chosen incident, presenting key metadata such as the incident ID, source device, risk level, Random Forest confidence score, and detection timestamp. The centerpiece of this view is a horizontal bar chart that visualizes the contribution of the top ten most influential features to the model's decision. Red bars indicate features that pushed the classification toward an attack verdict, while green bars represent features that pulled it toward a normal classification. Features highlighted in bold red denote anomalous values that had the strongest impact on the final decision. This level of transparency gives the security manager a scientifically grounded and fully auditable understanding of every decision made by the system.



Figure 6. SHAP Forensics — Explainable Attack Classification

This interface empowers the security manager to take direct, informed action against each detected incident in a structured and efficient manner. Active incidents are displayed as cards arranged in a two-column grid, each containing essential information including the source device name, incident timestamp, unique incident ID, potential production impact, and the system's recommended response. Each card provides four action buttons: Isolate, which immediately disconnects the affected device from the network to prevent further spread; Monitor, which places the device under continuous observation without interrupting its operation; Ignore, which classifies the incident as a false positive and dismisses it from the active queue; and Escalate, which forwards the incident to a higher level of authority for further investigation. This streamlined design significantly reduces response time and grants the manager full control over every threat individually, eliminating the need to navigate through complex log files.

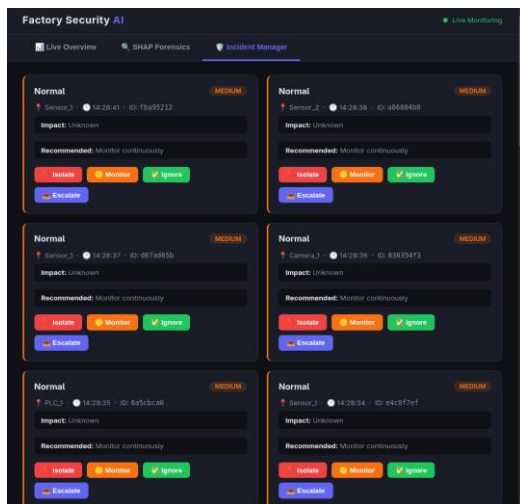


Figure 7. Incident Manager — Response Action Panel

9. SYSTEM ARCHITECTURE OVERVIEW

Table 2. Deployed Services — Docker Compose Configuration

Layer	Port	Model	Function
Simulator	—	—	Generates traffic (15% attacks, 85% normal)
Fog	5001	DT v3 + IsoForest	Binary filter: normal vs. suspicious
Cloud	5002	XGBoost + SHAP	Attack type (9 classes) + explanation
Dashboard	5003	—	Technical manager interface
Monitor	5004	—	Non-technical observer interface

10. APPLICATIONS & IMPACT

The system targets small-to-medium industrial facilities lacking dedicated SOC teams. By deploying Fog nodes co-located with factory floor switches, detection latency is reduced to under 200 ms without transmitting raw traffic off-site, preserving operational data confidentiality. The observer dashboard enables non-specialist staff (floor supervisors, shift managers) to act as a first line of awareness, reducing mean-time-to-notify by eliminating the need for technical interpretation before alerting security personnel.

The architecture is horizontally scalable: additional Fog nodes can be added per production line without modifying Cloud or Dashboard services. The bilingual interface

extends accessibility in multilingual workforces.

11. CHALLENGES & LIMITATIONS

The primary limitation is dataset scope: UNSW-NB15 does not include Modbus or MQTT-specific attack vectors, so detection of protocol-level OT attacks (e.g., Modbus function-code injection) relies on anomaly detection rather than supervised classification.

The XGBoost multi-class accuracy (76.15%) is constrained by inter-class similarity in UNSW-NB15, particularly between Exploits and Fuzzers. Future work will incorporate TON_IoT and SWaT datasets to improve OT-specific coverage.

SHAP computation adds approximately 15–25 ms per alert at the Cloud layer; under sustained high-rate attacks this may

introduce brief queue build-up, addressable by batching or async SHAP computation.

12. CONCLUSION

This work presents a deployable, explainable, and role-aware IDS for smart factories. The Fog–Cloud separation achieves both low-latency binary filtering (Recall 99.89%) and accurate multi-class classification (Accuracy 76.15%). Adversarial training and SMOTE oversampling harden the models against evasion and class imbalance respectively. SHAP integration closes the gap between model predictions and operator trust. Live system tests confirm practical viability with detection rates exceeding 97% under realistic mixed-traffic conditions. Future directions include integration of OT-specific datasets, WebSocket-based real-time streaming, and hardware deployment on Raspberry Pi-class edge nodes.

REFERENCES

- [1] Langner, R. (2011). Stuxnet: Dissecting a cyberwarfare weapon. *IEEE Security & Privacy*, 9(3), 49–51.
- [2] Shi, W., & Dustdar, S. (2016). The promise of edge computing. *Computer*, 49(5), 78–81.
- [3] Moustafa, N., & Slay, J. (2015). UNSW-NB15: A comprehensive data set for network intrusion detection systems. *Proc. MilCIS*, 1–6.
- [4] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proc. KDD*, 785–794.
- [5] Lundberg, S., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS*, 30.
- [6] Chawla, N. V., et al. (2002). SMOTE: Synthetic minority over-sampling technique. *JAIR*, 16, 321–357.
- [7] Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. *Proc. ICDM*, 413–422